

Week 4 - Individual - Class Exercise -- Graph Analysis Exercise – Amazon Network (Using Gephi)

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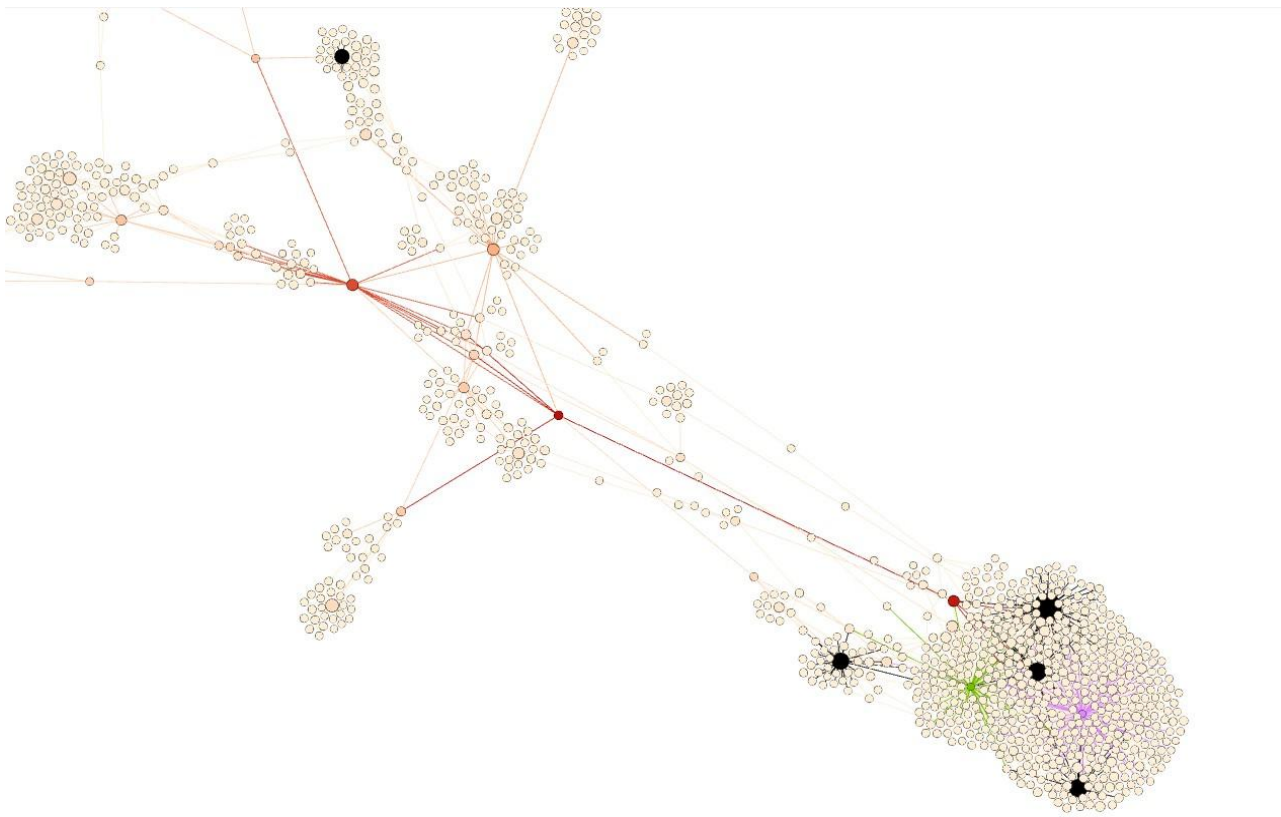
1. Introduction & Dataset Description

- **Dataset:** amazon_sampled_graph.csv.
- **Network Nature:** This is an **undirected graph** representing products that are frequently "co-purchased" on Amazon. Each node represents a product, and each edge represents a purchase link between two items.
- **Significance:** Understanding this network helps identify "Hub" products (popular items) and "Bridge" products (items that link different consumer interests), which is essential for recommendation engine optimization.

2. Graph Visualization & Layout

For this analysis, the **ForceAtlas2** layout was applied. This layout is highly effective for organic structures as it uses a linear-linear attraction and repulsion model to group related nodes while pushing apart unrelated ones, making communities visually distinct.

- **Adjustments:** Node sizes were scaled based on **Degree** to highlight popular products, and edge thickness was adjusted to improve the overall clarity of the co-purchasing paths.



3. Degree Analysis

The **Degree** metric indicates the total number of direct co-purchasing connections a product has. The top 5 nodes with the highest degree are the most popular "Hubs" in the network:

1. **Node 310** (Highest connections)
2. **Node 1684**
3. **Node 1912**
4. **Node 3437**
5. **Node 0**

4. Centrality Measures & Influence

To find the most "influential" products rather than just the most popular ones, we computed **Betweenness Centrality** and **PageRank**.

Top 5 Influential Nodes (PageRank)

PageRank determines the importance of a node based on its connections to other important nodes.

- Nodes **310**, **1684**, **1912**, **0**, and **3437** consistently ranked highest, showing they are not just connected to many products, but to other central products in the Amazon catalogue.

Top 5 Bridge Nodes (Betweenness Centrality)

Betweenness identifies nodes that act as "bridges" between different product groups.

- **Node 107** and **Node 1085** appeared as significant bridges. If these products were removed, the connectivity between different consumer communities would significantly weaken.

5. Community Detection

The **Modularity** algorithm was used to detect distinct communities within the network.

- **Insights:** By colouring nodes based on their **Modularity Class**, we identified clusters of products that are almost always bought together. These likely represent specific departments (e.g., Electronics vs. Books) or niche consumer hobbies.

6. Filtering & Highlighting

To provide a clear audit of the network's core, the following filters were applied:

- **Degree Filter:** Isolated nodes with a **Degree** ≥ 5 to remove "leaf" products (items bought with only one or two others) and reveal the network's social backbone.
- **Emphasis:** The top 5 nodes by degree were manually scaled to a larger size and coloured **Black**. This visually confirms their status as the dominant anchors of the purchasing network.

7. Conclusion & Final Insights

- **Hub Products:** High-degree nodes (colored black) are the main drivers of traffic and should be the primary focus for marketing placements.
- **Cross-Departmental Linking:** Products with high **Betweenness Centrality** are vital for cross-selling, as they link different consumer communities.
- **Data Integrity:** By cleaning the graph and removing duplicates/outliers during the profiling phase, we ensured that these metrics accurately reflect real-world consumer behavior rather than data noise.